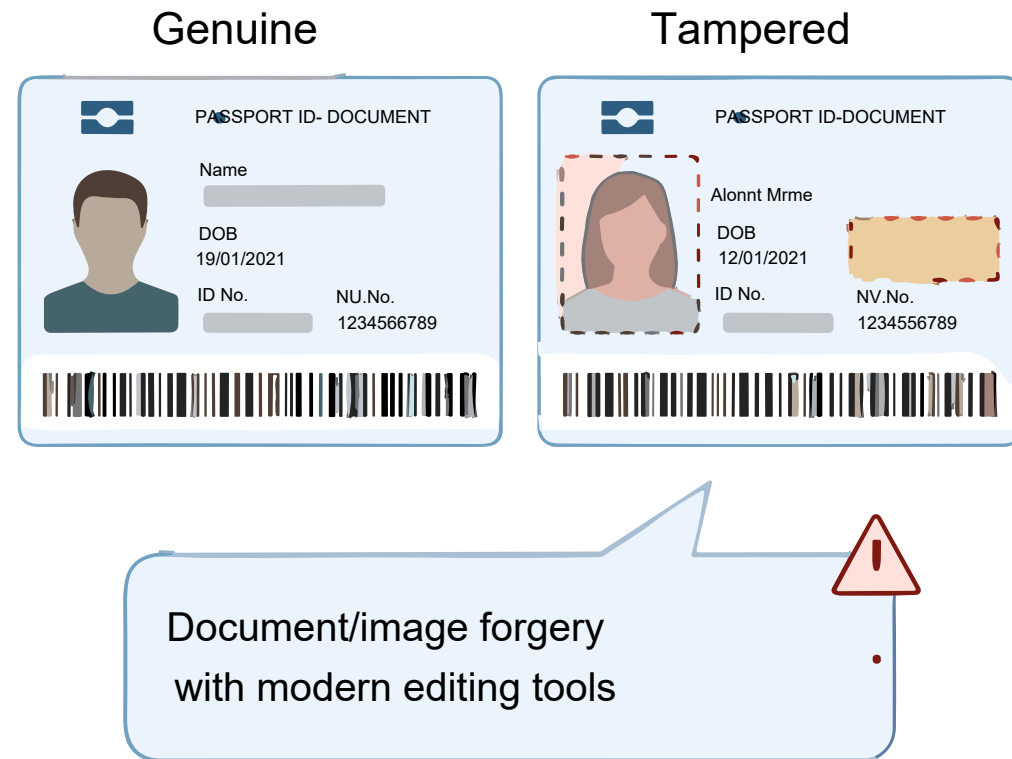
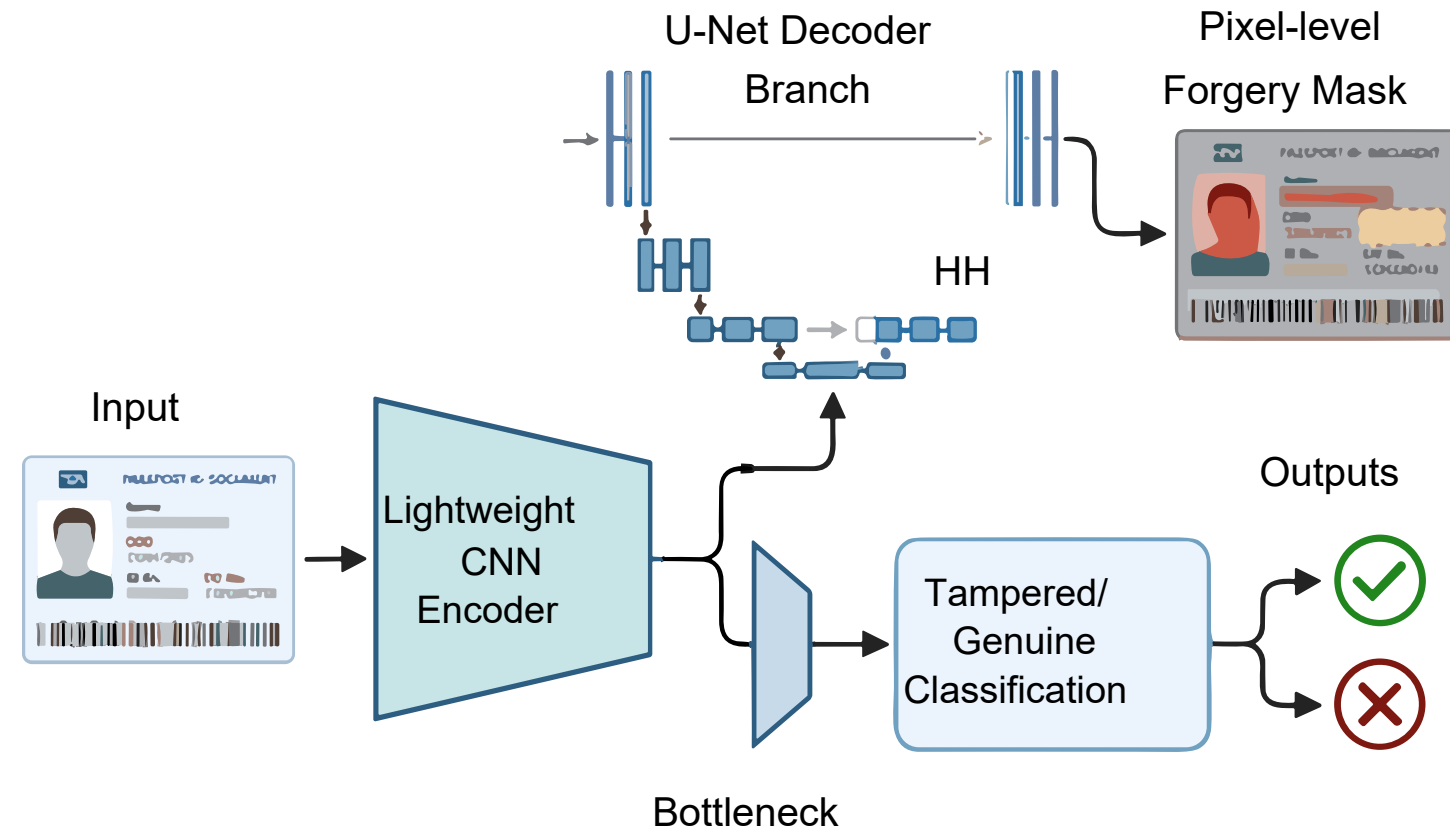


Multi-objective Pareto hyperparameter optimization for joint detection and localization of forged identity documents

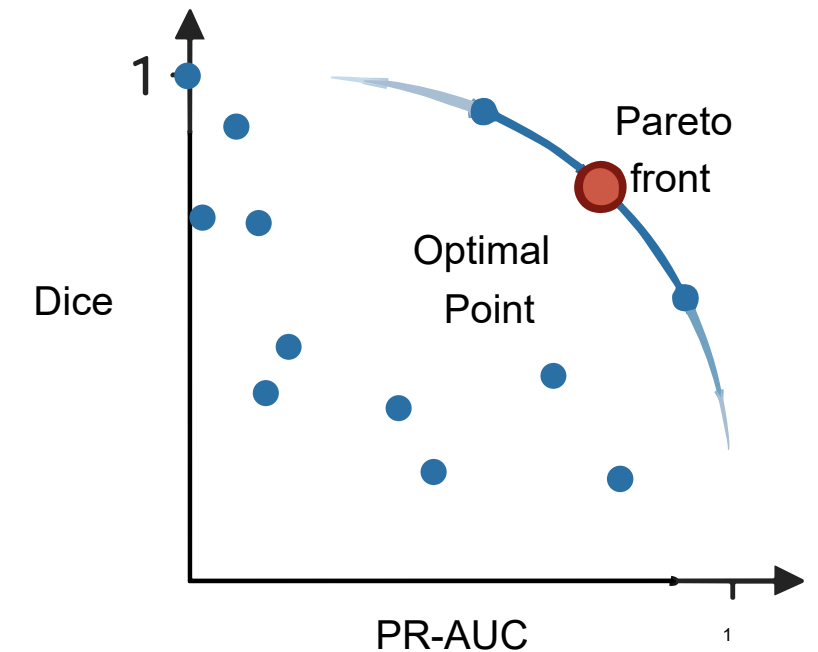
Problem Illustration



Solution Overview

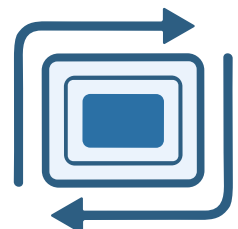


Optimization Approach



Multi-objective HPO: PR-AUC & Dice optimized jointly, no manual weighting

Evaluation



10×5 Nested CV, MOTPE 2,500 Trials



Blind holdout test



Wilcoxon test and Holm-Bonferroni correction